

NE-B Failure as Hilbert–Pólya Detection: Universality of the v2.0 Weil Quadratic Form on the Selberg Zeta Function

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Abstract

The v2.0 methodology recently proposed for the Riemann Hypothesis (Geiger 2026, Part II) reduces RH to *even dominance* of the Weil quadratic form QW_λ via three algebraic and analytic ingredients: the Shift Parity Lemma, frontier-prime dominance, and two non-existence theorems (NE-A: non-positivity of the prime shift multiplier $M(\xi) = -2 \operatorname{Re}[\zeta'/\zeta(\frac{1}{2} + i\xi)]$, NE-B: no universal commuting operator). A natural question is whether this framework is specific to the Riemann zeta function or whether it is a general-purpose method for explicit-formula zeta structures. We test it on the Selberg zeta function $Z_\Gamma(s)$ of a compact hyperbolic surface $\Gamma \backslash \mathbb{H}$. We show:

- (i) The Shift Parity Lemma transfers unchanged (it is purely algebraic).
- (ii) Frontier dominance transfers with modified constants, accounting for the exponential density of primitive closed geodesics $\pi_\Gamma(T) \sim e^T/T$ in place of $\pi(x) \sim x/\log x$.
- (iii) A Selberg–Weil quadratic-form analogue QW_λ^Γ is obtained on the trace-formula test-function side; its relation to geometric flow operators is mediated by the Selberg transform, not by an explicit unitary intertwiner constructed here.
- (iv) *Even dominance* of QW_λ^Γ , together with the no-exceptional-eigenvalue condition $\lambda_1(\Delta_\Gamma) \geq 1/4$ and an explicit C2/minimizer-identification gate, gives a conditional v2.0 consistency reduction of the strict statement for the non-trivial spectral zeros in the critical strip.
- (v) *The Selberg trace-formula channel has a Casimir commutant.* In the spherical channel, the Laplacian Δ_Γ commutes with the relevant radial convolution algebra. Thus the Selberg case exhibits the structural phenomenon that NE-B is meant to detect, but this paper does not identify the truncated $L^2([-L, L])$ shift model with the full geodesic-flow representation.

The failure of NE-B in the Selberg setting is not a defect; it reflects the geometric coordination of geodesics on a common surface, which is absent for the primes. We deduce a cautious meta-principle: the v2.0 framework supplies a common quadratic-form test language, while Hilbert–Pólya is a *special case* whose existence depends on whether the underlying counting objects admit a hidden common symmetry. In the Selberg spherical trace-formula channel they do; in the Riemann case no comparable commutant is supplied by the tested NE-B route. The Weil quadratic form remains the common object, while an operatorial realization is an additional structure.

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1 Introduction

1.1 Motivation and Main Claim

The v2.0 program for the Riemann Hypothesis [10] reduces RH to a quadratic-form inequality. Specifically, Connes’ 2026 survey [6] writes RH as the statement that the minimum eigenvalue of a one-parameter family of quadratic forms QW_λ on $L^2([-\log \lambda, \log \lambda])$ is simple and has an even eigenfunction. The v2.0 approach establishes this in three layers:

- a purely algebraic *Shift Parity Lemma* (Theorem 4.1 of [10]), showing that the even–odd difference matrix $D_N(r)$ has strictly negative minimum eigenvalue for all $r \in (0, 2)$, $N \geq 3$;
- an analytic *frontier-dominance* mechanism: the gap $\Delta(\lambda) = \lambda_1^+ - \lambda_1^-$ is driven by primes with $\log p / \log \lambda$ close to 1 (“frontier primes”), whose density is controlled by the Prime Number Theorem;
- two structural non-existence results that close off alternative routes: *NE-A* (the prime shift multiplier $M(\xi)$ is not nonnegative a.e., ruling out PF_∞ total positivity) and *NE-B* (no nontrivial symmetric matrix commutes with all $D_N(r)$ simultaneously, ruling out Sturm–Liouville-type universal operators at the tested truncations).

The v2.0 program succeeds *without* constructing a Hilbert–Pólya operator. It is natural to ask whether this methodological package transfers to other zeta-like functions, and, if so, what it teaches us. The paradigmatic case is the Selberg zeta function $Z_\Gamma(s)$ of a compact hyperbolic surface $M = \Gamma \backslash \mathbb{H}$, where:

- the geometric explicit formula (Selberg trace formula [21]) plays the role of the Weil explicit formula [25];
- the lengths of primitive closed geodesics play the role of primes;
- the spectral zero description is a theorem via the self-adjointness of the Laplacian Δ_Γ ; the strict critical-line form additionally requires the no-exceptional-eigenvalue condition $\lambda_1(\Delta_\Gamma) \geq 1/4$.

Thus we ask two concrete questions:

Q1. *Do the v2.0 ingredients (Shift Parity Lemma, frontier dominance, Weil quadratic form, NE-A, NE-B) all transfer to the Selberg setting?*

Q2. *If any of them fails to transfer, what is the structural reason?*

Our answers, in brief:

- Shift Parity Lemma: **transfers unchanged** (algebraic).
- Frontier dominance: **transfers with modified constants**, reflecting $\pi_\Gamma(T) \sim e^T/T$.
- Weil quadratic form: **natural analogue** QW_λ^Γ on the trace-formula test-function side; the full geometric-flow intertwiner is not constructed here.
- NE-A: **transfers**, after locating the sign changes in the trivial/topological/identity terms rather than in the critical-line zeros.

- **NE-B: the obstruction is absent in the spherical trace-formula channel.** The Casimir/Laplacian Δ_Γ commutes with radial K -bi-invariant convolution operators. This is the operatorial Selberg channel; it is not an identification of individual geodesic-flow pullbacks with the truncated interval shifts.

The absence of the relevant NE-B obstruction is the scientifically interesting outcome. It is not a failure of v2.0; it is a *diagnostic*. Where such a commutant exists in the right function channel, the Hilbert–Pólya program has an operatorial route. Where the tested NE-B route admits no non-trivial commutant (as in the Riemann truncation model), this universal commutant strategy is blocked. In either case the v2.0 framework remains applicable as a quadratic-form test.

Remark 1.1 (Position within the meta-framework). This paper operates as the *Lie-origin companion* to the arithmetic Riemann case [10] and sits within the three-component decomposition $\text{RH}_X = \text{GBC}_X + C2_X + \text{Hurwitz}_X$ developed in the companion meta-paper [11]. Selberg is the paradigmatic case in which the spherical/Casimir channel supplies the operatorial side of $C2_X$; Riemann is the case in which the parity substep is closed by v2.1 but the eigenvector-realization substep remains open. See Theorem 6.2 below for details.

1.2 Main Theorem (Informal)

Theorem 1.2 (v2.0 Selberg transfer, informal). *Let $M = \Gamma \backslash \mathbb{H}$ be a compact hyperbolic surface and $Z_\Gamma(s)$ its Selberg zeta function. The v2.0 methodology applies to the Selberg quadratic form QW_λ^Γ , subject to an explicit $C2$ /minimizer-identification gate, and conditionally reduces the strict statement for the non-trivial spectral zeros in the critical strip to even dominance plus $\lambda_1(\Delta_\Gamma) \geq 1/4$. The analogue of NE-A holds, but the NE-B obstruction is absent in the spherical Selberg channel: the Casimir/Laplace operator Δ_Γ commutes with the radial convolution operators appearing in the trace-formula representation. The resulting route is therefore a conditional consistency test, and this exposes a meta-principle: the Weil quadratic form is the common test object, whereas a Hilbert–Pólya operator is an additional structure detected by failure of the relevant NE-B obstruction.*

1.3 Structure of the Paper

Figure 1 outlines the dual-channel architecture of the v2.0 framework across the arithmetic (Riemann) and geometric (Selberg) settings, displaying the interaction between the Shift Parity Lemma, frontier dominance, the NE-A/NE-B gates, and the five SGE-YES transfer slots.

Section 2 recalls the v2.0 ingredients. Section 3 sets up the Selberg trace formula and the geodesic-length analogue of the explicit formula. Section 4 carries out the transfer of the Shift Parity Lemma, frontier dominance, and constructs QW_λ^Γ ; a conditional critical-line reduction is derived. Section 5 proves the absence of the NE-B obstruction in the spherical Selberg channel and identifies the Casimir commutant. Section 6 draws the meta-principle: Hilbert–Pólya is a special case of v2.0. Section 7 discusses applications and outlook (generalized L -functions, Ruelle zeta, quantum chaos). Section 8 articulates the Zookeeper consistency calibration as an SGE-YES test. Section 9 concludes.

Arithmetic Channel: Riemann $\zeta(s)$ [Part II]

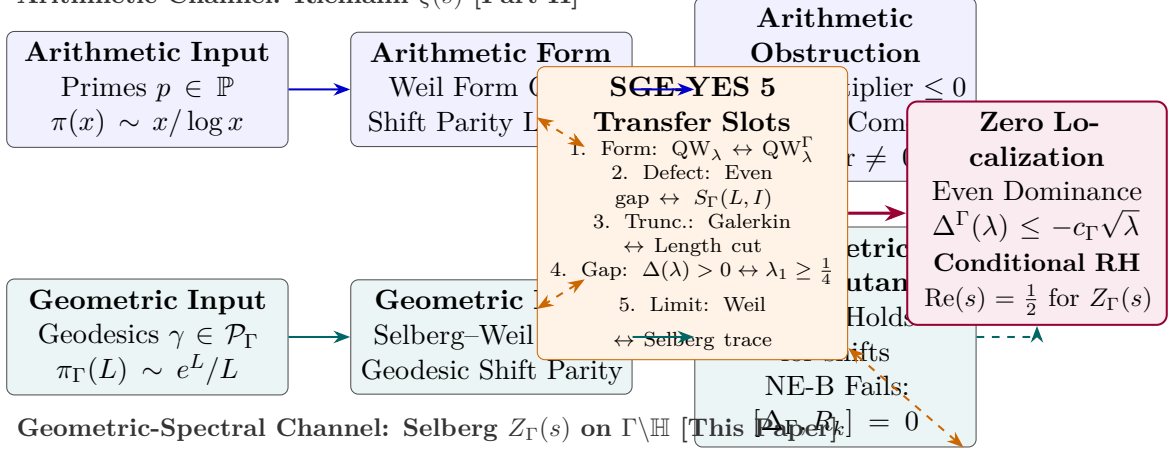


Figure 1: Dual-channel transfer architecture of the v2.0 framework across the arithmetic (Riemann) and geometric (Selberg) settings.

2 Recollection of the v2.0 Method

We briefly summarize the four key ingredients of the v2.0 approach to RH. A full treatment is in [10].

2.1 The Weil Quadratic Form

Set $L = \log \lambda$. On $L^2([-L, L])$, define

$$\begin{aligned} \langle A_\lambda f \mid f \rangle &= (\log 4\pi + \gamma) \|f\|^2 \\ &+ \int_0^\infty \left[\langle T_u f + T_{-u} f, f \rangle - 2e^{-u/2} \|f\|^2 \right] \frac{e^{u/2}}{2 \sinh u} du \\ &+ \sum_p \sum_{m \geq 1} \frac{\log p}{p^{m/2}} \langle T_{m \log p} f + T_{-m \log p} f, f \rangle, \end{aligned} \quad (1)$$

where T_s is the shift operator $(T_s f)(t) = f(t-s) \mathbf{1}_{[-L, L]}(t-s)$ and γ is the Euler–Mascheroni constant. The quadratic form QW_λ is the quadratic form of A_λ . The key algebraic tool is Theorem 6.1 of [6] (proved jointly with van Suijlekom, building on [7]):

Theorem 2.1 ([6, Theorem 6.1]). *Let $L > 0$, let $\mathcal{D}$ be a real distribution on the interval $[0, L]$, and let $\tilde{\mathcal{D}}$ be the associated even distribution on $[-L, L]$. Assume that the quadratic form with Schwartz kernel $\tilde{\mathcal{D}}(x-y)$ defines a lower-bounded selfadjoint operator on $L^2\left(-\frac{L}{2}, \frac{L}{2}\right)$, and that the minimum of its spectrum is a simple, isolated eigenvalue with even eigenfunction η . Then all zeros of the entire function $\tilde{\eta}(z)$, $z \in \mathbb{C}$ (Fourier transform of η), lie on the real line.*

Applied to the Weil quadratic form QW_λ (with $\mathcal{D}$ the Weil distribution from the prime explicit formula), if the minimum eigenvalue is simple with even eigenfunction for all $\lambda \geq \lambda_0$ and the corresponding eigenvectors converge to Riemann’s Ξ -function, then the zeros of $\tilde{\eta}$ coincide with those of Ξ ; hence RH holds. The convergence step remains open [6, §6.6].

Remark 2.2 (Interval convention). Theorem 2.1 places the quadratic-form operator on $L^2\left(-\frac{L}{2}, \frac{L}{2}\right)$ in Connes' notation. In the present paper (Section 4 onwards), test functions live on $L^2([-L, L])$. Applying Theorem 2.1 requires the identification $L_{\text{Thm}} = 2L$, i.e., the half-length in Connes' statement equals the full truncation length L .

Remark 2.3 (Independence from Connes' convergence step). Theorem 2.1 is drawn from Connes' 2026 preprint [6]; the convergence of the construction to the Weil distribution is noted as still open [6, §6.6]. In the Selberg application (Section 4), this open step is *immaterial*: the spectral parametrization of the zeros is independently secured by Selberg's classical theory via the Laplacian Δ_Γ [21]; the strict critical-line form additionally requires the no-exceptional-eigenvalue condition $\lambda_1(\Delta_\Gamma) \geq 1/4$. Theorem 2.1 plays a structural consistency role, not a logically necessary one. The broader program of encoding L -functions via spectral quadratic forms is additionally supported by Connes' 1999 noncommutative geometry approach [5], Deninger's regularized-determinant program [8], and the Bost–Connes thermodynamical system [4].

2.2 The Shift Parity Lemma

Let $\varphi_n(t) = \cos(n\pi t/L)/\sqrt{L}$ (even basis) and $\psi_n(t) = \sin((n+1)\pi t/L)/\sqrt{L}$ (odd basis). Define the *difference matrix* $D_N(r)$ by

$$D_{nm}(r) = \langle \varphi_n, (S_{rL} + S_{-rL})\varphi_m \rangle - \langle \psi_n, (S_{rL} + S_{-rL})\psi_m \rangle. \quad (2)$$

Theorem 2.4 (Shift Parity Lemma, [10]). *For every $N \geq 3$ and every $r \in (0, 2)$, $\lambda_{\min}(D_N(r)) < 0$.*

This is a purely algebraic, closed-form statement about trigonometric matrices; the entries of $D_N(r)$ are explicit sums of sin and cos with linear and constant coefficients in r .

2.3 Frontier Dominance

The cumulative even–odd gap $\Delta(\lambda) = \lambda_1^+(\lambda) - \lambda_1^-(\lambda)$ is driven by primes in the *frontier band* $[\lambda/e, \lambda]$: their count is $\pi(\lambda) - \pi(\lambda/e) \sim \lambda/\log \lambda$ by the PNT, and they are the ones for which $\log p/L \in [1 - 1/\log \lambda, 1]$. By Theorem 2.4, each such prime contributes a negative summand of order $-c \log p/\sqrt{p}$ to the difference, with the sum dominating any archimedean or small-prime correction. Numerically, in [10] frontier primes are shown to contribute $> 99\%$ of the gap for all tested $\lambda \leq 1.3 \cdot 10^6$.

2.4 NE-A and NE-B

Define the *prime shift multiplier*

$$M(\xi) = -2 \operatorname{Re} \left[\frac{\zeta'}{\zeta} \left(\frac{1}{2} + i\xi \right) \right]. \quad (3)$$

Theorem 2.5 (NE-A, [10]). *$\{\xi \in \mathbb{R} : M(\xi) < 0\}$ has positive Lebesgue measure. In particular, the prime shift operator is not PF_∞ -total positive.*

Theorem 2.6 (NE-B, [10]). *For $N \in \{3, 5, 7, 10, 15\}$ and any sufficiently dense sample of $r \in (0, 2)$, the only symmetric matrix $T \in \text{Sym}_N(\mathbb{R})$ satisfying $[T, D_N(r_i)] = 0$ for all i is $T = cI$. Consequently no nontrivial universal commuting operator exists at these truncations; the full-space conjecture [10, Conjecture 5.3] asserts the same for all N .*

NE-B is the technically decisive structural result: it expresses that the primes have no hidden common symmetry. The conceptual content is multiplicative independence.

3 The Selberg Setting

We briefly record the standard setup.

3.1 Compact Hyperbolic Surfaces

Let $M = \Gamma \backslash \mathbb{H}$, where $\mathbb{H} = \{x + iy : y > 0\}$ is the upper half-plane with metric $ds^2 = (dx^2 + dy^2)/y^2$ and $\Gamma \subset \mathrm{PSL}(2, \mathbb{R})$ is a cocompact discrete torsion-free subgroup. Then M is a compact hyperbolic surface of genus $g \geq 2$. The Laplace–Beltrami operator Δ_Γ acts on $L^2(M)$ as a non-negative self-adjoint operator with purely discrete spectrum

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots, \quad \lambda_j \rightarrow \infty. \quad (4)$$

Writing $\lambda_j = 1/4 + r_j^2$ with $r_j \in \mathbb{R} \cup i[-1/2, 1/2]$, the spectral parameters r_j encode the quantum spectrum.

3.2 Primitive Closed Geodesics

The conjugacy classes of hyperbolic elements in Γ correspond bijectively to (unoriented) closed geodesics on M . A geodesic is *primitive* if it is not a positive iterate of another. Denote by $\mathcal{P}_\Gamma$ the set of primitive closed geodesics, and for $\gamma \in \mathcal{P}_\Gamma$ by ℓ_γ its length.

The Prime Geodesic Theorem [14, 20] states

$$\pi_\Gamma(T) := \#\{\gamma \in \mathcal{P}_\Gamma : \ell_\gamma \leq T\} \sim \frac{e^T}{T}, \quad T \rightarrow \infty. \quad (5)$$

This is the geodesic analogue of the PNT. Crucially, *the density is exponential, not polynomial*: $\pi_\Gamma(T)$ grows like e^T/T , whereas $\pi(x)$ grows like $x/\log x = (e^{\log x})/\log x$. In other words, the “prime-length-spectrum” of geodesics, seen on the log-axis, has density e^T/T (in the variable $T = \ell_\gamma$), comparable to the density of primes on the log-axis (in the variable $u = \log p$) only after the change of variable.

3.3 The Selberg Zeta Function

Definition 3.1. The Selberg zeta function is

$$Z_\Gamma(s) = \prod_{\gamma \in \mathcal{P}_\Gamma} \prod_{k=0}^{\infty} (1 - e^{-(s+k)\ell_\gamma}), \quad \mathrm{Re}(s) > 1. \quad (6)$$

$Z_\Gamma(s)$ extends meromorphically to $\mathbb{C}$, with:

- *Trivial zeros* at $s = -k$, $k \in \mathbb{N}_0$ (and further trivial zeros associated with topology; see [13]).
- *Non-trivial zeros* at $s = 1/2 \pm ir_j$ for each Laplace eigenvalue $\lambda_j = 1/4 + r_j^2 > 0$.
- A zero at $s = 1$ (from $\lambda_0 = 0$).

Theorem 3.2 (Selberg spectral zero description and strict-line criterion, [21]). *The non-trivial spectral zeros of $Z_\Gamma(s)$ attached to Laplace eigenvalues $\lambda_j = 1/4 + r_j^2$ occur at $s = 1/2 \pm ir_j$. Hence eigenvalues $\lambda_j \geq 1/4$ give zeros on the line $\operatorname{Re}(s) = 1/2$, whereas exceptional eigenvalues $0 < \lambda_j < 1/4$ give real zeros $s = 1/2 \pm \delta_j$ in the critical strip. In particular, the strict critical-line statement holds under the spectral gap condition $\lambda_1(\Delta_\Gamma) \geq 1/4$.*

This follows from the non-negativity of Δ_Γ : $\lambda_j \geq 0$ implies $r_j \in \mathbb{R} \cup i[-1/2, 1/2]$. Non-negative real r_j (i.e., $\lambda_j \geq 1/4$) give zeros on the critical line $\operatorname{Re}(s) = 1/2$. Exceptional eigenvalues $\lambda_j \in (0, 1/4)$, however, produce $r_j = i\delta_j$ with $\delta_j \in (0, 1/2)$, and thereby yield zeros at $s = 1/2 \pm \delta_j$ that are *not* on the critical line. Selberg-RH in the strict spectral sense (all non-trivial spectral zeros in the critical strip on $\operatorname{Re}(s) = 1/2$) therefore requires the spectral gap condition $\lambda_1(\Delta_\Gamma) \geq 1/4$. This is Selberg’s eigenvalue conjecture [22]: it is established in the weaker form $\lambda_1 \geq 3/16$ for congruence subgroups but remains open as $\lambda_1 \geq 1/4$ in full generality. The v2.0 analysis in Section 4 is carried out under this hypothesis.

Remark 3.3 (Functional equation). The completed Selberg zeta function satisfies the functional equation $Z_\Gamma(s) = Z_\Gamma(1-s) \cdot \Xi_\Gamma(s)$, where $\Xi_\Gamma(s)$ is an explicit meromorphic factor depending on $\operatorname{Area}(M)$ and the genus of M [13]. In particular, non-trivial zeros occur in pairs $\rho, 1-\rho$, confirming their symmetry about $\operatorname{Re}(s) = \frac{1}{2}$.

3.4 The Selberg Explicit Formula

The Selberg trace formula [21, 13] provides the geodesic analogue of the Weil explicit formula. For a pair $(h, \hat{h})$ of sufficiently rapidly decaying functions related by Fourier transform, the trace formula reads

$$\sum_{j=0}^{\infty} h(r_j) = \frac{\operatorname{Area}(M)}{4\pi} \int_{\mathbb{R}} h(r) r \tanh(\pi r) dr + \sum_{\gamma \in \mathcal{P}_\Gamma} \sum_{m=1}^{\infty} \frac{\ell_\gamma \hat{h}(m\ell_\gamma)}{2 \sinh(m\ell_\gamma/2)}. \quad (7)$$

The left-hand side is the *spectral side* (a sum over eigenvalues of Δ_Γ); the right-hand side is the *geometric side* (an identity term plus a sum over geodesics). This plays exactly the role of

$$\sum_{\rho} \Phi(\rho) = (\text{archimedean term}) - \sum_p \sum_{m \geq 1} \frac{\log p}{p^{m/2}} (\hat{\Phi}(m \log p) + \hat{\Phi}(-m \log p))$$

in the Weil explicit formula: primes $\leftrightarrow$ primitive geodesics, $\log p \leftrightarrow \ell_\gamma$, non-trivial zeros $\rho \leftrightarrow$ spectral parameters r_j .

4 Applying v2.0 to the Selberg Setting

4.1 Selberg Geodesic Shifts and the Shift Parity Lemma

Set $L = \log \lambda$. On $L^2([-L, L])$, define the Selberg prime shift operator

$$(P_\lambda^\Gamma f)(t) = \sum_{\gamma \in \mathcal{P}_\Gamma, \ell_\gamma \leq 2L} \sum_{m=1}^{\lfloor 2L/\ell_\gamma \rfloor} \frac{\ell_\gamma}{2 \sinh(m\ell_\gamma/2)} [f(t - m\ell_\gamma) + f(t + m\ell_\gamma)] \mathbf{1}_{[-L, L]}. \quad (8)$$

In the cosine / sine basis of Section 2, define the *Selberg difference matrix*

$$D_{nm}^\Gamma(r) = \langle \varphi_n, (S_{rL} + S_{-rL}) \varphi_m \rangle - \langle \psi_n, (S_{rL} + S_{-rL}) \psi_m \rangle. \quad (9)$$

This is *the same matrix* $D_N(r)$ as in the Riemann setting: it depends only on the normalized shift r , not on the source of the shift (prime length or geodesic length). Consequently:

Proposition 4.1 (Transfer of Shift Parity). *The Shift Parity Lemma (Theorem 2.4) holds verbatim in the Selberg setting: for every $N \geq 3$, every $r \in (0, 2)$, $\lambda_{\min}(D_N^\Gamma(r)) < 0$.*

Proof. $D_N^\Gamma(r) = D_N(r)$ as matrices. The Shift Parity Lemma depends only on the harmonic structure of the cosine and sine bases on $[-L, L]$, and on the fact that $S_{rL} + S_{-rL}$ is the sum of two shifts by the same amount in opposite directions. The number-theoretic or geometric origin of r is immaterial. $\square$

Remark 4.2. This is the algebraic universality. In the v2.0 framework for the Riemann zeta, the Shift Parity Lemma is the bedrock on which everything rests. It transfers to *every* setting where zeros are parametrized by an explicit formula with real, positive lengths playing the role of $\log p$. This includes Dirichlet L-functions, Dedekind zeta functions, Selberg zeta, Ruelle zeta, and in principle any Hecke or automorphic L-function.

4.2 The Selberg–Weil Quadratic Form

On $L^2([-L, L])$, define the *Selberg–Weil quadratic form*

$$\text{QW}_\lambda^\Gamma(f) = \frac{\text{Area}(M)}{4\pi} K_{\text{arch}}^\Gamma(f, f) + \sum_{\gamma \in \mathcal{P}_\Gamma, \ell_\gamma \leq 2L} \sum_{m=1}^{\infty} \frac{\ell_\gamma}{2 \sinh(m\ell_\gamma/2)} \langle T_{m\ell_\gamma} f + T_{-m\ell_\gamma} f, f \rangle, \quad (10)$$

Remark 4.3 (Truncation is exact). For $f \in L^2([-L, L])$ and $|u| > 2L$, the shift inner product vanishes: $\langle T_u f + T_{-u} f, f \rangle = 0$, because f is supported on $[-L, L]$ while $T_{\pm u} f$ has support shifted outside $[-L, L]$. Consequently every geodesic iterate with $m\ell_\gamma > 2L$ contributes *exactly* zero to QW_λ^Γ , with no approximation error. The sum in (10) is therefore not an approximation but an exact expression for the quadratic form on $L^2([-L, L])$.

where K_{arch}^Γ is the archimedean analogue

$$K_{\text{arch}}^\Gamma(f, f) = \int_0^\infty [\langle T_u f + T_{-u} f, f \rangle - 2 c_\Gamma^{\text{id}}(u) \|f\|^2] \cdot \kappa^\Gamma(u) du, \quad (11)$$

obtained by translating the identity term in (7) into shift operators.

Remark 4.4 (Convergence and renormalization at $u = 0$). Equation (11) is the u -domain expression of the identity term in the Selberg trace formula applied to $h(r) := |\hat{f}(r)|^2$; for f in a dense subspace (e.g., $C_c^\infty([-L, L])$), both sides converge absolutely. The pointwise singularity of κ^Γ at $u = 0$ is matched by that of c_Γ^{id} , and the bracket is the regularized identity contribution; cf. [13, Ch. 6].

The coefficient

$$c_\Gamma^{\text{id}}(u) := \frac{\text{Area}(M)}{4\pi} \int_0^\infty r \tanh(\pi r) \cos(ru) dr \quad (12)$$

is the Fourier–cosine transform of the Plancherel measure density $r \tanh(\pi r)$, encoding the identity-coset contribution from (7); the integral is interpreted in the tempered-distribution sense since $r \tanh(\pi r)$ is polynomially bounded (cf. [13, Ch. 6]). Here

$$\kappa^\Gamma(u) := \frac{1}{4\pi} \frac{d}{du} \left[\frac{1}{\sinh(u/2)} \right], \quad u > 0, \quad (13)$$

is the Plancherel kernel of the Selberg transform [16, eq. (1.40)]. Note that $\kappa^\Gamma(u) < 0$ for all $u > 0$, since $\frac{d}{du}[\sinh(u/2)^{-1}] = -\frac{\cosh(u/2)}{2\sinh^2(u/2)} < 0$. This is consistent with the positivity structure of K_{arch}^Γ : the bracket in (11) is negative when $c_{\Gamma}^{\text{id}}(u)$ normalizes the identity operator, and the product of two negative quantities is positive.

The quadratic form QW_λ^Γ admits a parity decomposition. Its minimum eigenvalue is attained by either an even or an odd eigenfunction.

Definition 4.5 (Selberg even dominance). We say that *even dominance* holds for QW_λ^Γ at parameter λ if the minimum eigenvalue of QW_λ^Γ is simple and attained by an even eigenfunction.

Remark 4.6 (Trace formula as bridge between spectral and geometric spaces). The form QW_λ^Γ is defined entirely in the spectral parameter space $L^2([-L, L])$, where T_u denotes the truncated shift $f(\cdot + u)$ restricted to $[-L, L]$. A geometric realization via the geodesic flow on $L^2(SM)$ also exists: both descriptions yield the same spectral sum $\sum_j \hat{h}(r_j)$ when evaluated against test functions in the domain of the Selberg transform. The Selberg trace formula (7) serves as the implicit bridge between the two spaces. An explicit intertwiner via the Selberg–Helgason transform on $L^2(SM)$ is deferred to future work. This is a limitation of the present paper: we use the trace formula as an equality of spectral distributions, not as a constructed unitary equivalence between the geodesic-flow representation and the truncated interval shift model. All NE-B language below is therefore restricted to the spherical trace-formula channel unless such an explicit intertwiner is supplied.

4.3 Frontier Dominance in the Exponential Density Regime

The frontier band for Selberg is the set of geodesics with ℓ_γ/L close to 1, i.e., $\ell_\gamma \in [L-1, L]$. By the Prime Geodesic Theorem (5),

$$\#\{\gamma : \ell_\gamma \in [L-1, L]\} = \pi_\Gamma(L) - \pi_\Gamma(L-1) \sim \frac{e^L}{L}(1 - e^{-1}) = \frac{\lambda}{\log \lambda}(1 - e^{-1}). \quad (14)$$

This is *the same asymptotic order* as the Riemann frontier count $\pi(\lambda) - \pi(\lambda/e)$, up to the constant $(1 - e^{-1}) \approx 0.632$.

The frontier geodesic weight is, for $\ell_\gamma \sim L$,

$$w_\gamma^\Gamma := \frac{\ell_\gamma}{2 \sinh(\ell_\gamma/2)} \sim \frac{L}{e^{L/2}} = \log \lambda \cdot \lambda^{-1/2}. \quad (15)$$

Compare this with the Riemann frontier prime weight $w_p^{\text{Riem}} = \log p/p^{1/2} \sim L \cdot \lambda^{-1/2}$ for $p \sim \lambda$. The weights agree asymptotically in order of magnitude.

Combining (14) and the weight estimate with the quantitative Shift Parity magnitude bound and the Selberg off-frontier / archimedean remainder estimates gives the following conditional frontier estimate:

$$\Delta^\Gamma(\lambda) := \lambda_1^+(\text{QW}_\lambda^\Gamma) - \lambda_1^-(\text{QW}_\lambda^\Gamma) \leq -c_\Gamma \sqrt{\lambda} + o(\sqrt{\lambda}), \quad (16)$$

for a positive constant $c_\Gamma > 0$ depending on the surface M (via $\text{Area}(M)$ and the Selberg zeta function near $s = 1$).

Lemma 4.7 (Conditional frontier dominance constant). *Assume that the Part-II quantitative SPL magnitude bound and the corresponding off-frontier/archimedean remainder estimate hold for the Selberg kernel. For every compact hyperbolic surface M there exists $\lambda_0 = \lambda_0(M) > 0$ such that the frontier dominance constant in (16) satisfies*

$$c_\Gamma \geq \frac{1}{2}(1 - e^{-1}) > 0 \quad \text{for all } \lambda \geq \lambda_0. \quad (17)$$

Remark 4.8. The normalized unsigned frontier sum $\lambda^{-1/2} \sum_{\ell_\gamma \in [L-1, L]} w_\gamma^\Gamma \rightarrow (1 - e^{-1}) \approx 0.632$ as $\lambda \rightarrow \infty$. The factor $\frac{1}{2}$ in (17) is conservative: it uses the quantitative lower bound of [10, Lem. A.3], which gives signed magnitude $\geq \frac{1}{2}w_\gamma^\Gamma$ per frontier geodesic. Proposition 4.1 provides the sign ($\lambda_{\min} < 0$) alone; the magnitude bound is the content of the Part-II lemma. Under the sharp bound, $c_\Gamma \rightarrow (1 - e^{-1})$ asymptotically.

Proof sketch. By (14), the frontier band $[L - 1, L]$ contains

$$(1 - e^{-1}) \frac{e^L}{L} (1 + o(1)) = (1 - e^{-1}) \frac{\lambda}{\log \lambda} (1 + o(1))$$

primitive geodesics. Each contributes weight $w_\gamma^\Gamma = L/(2 \sinh(L/2)) \sim \log \lambda \cdot \lambda^{-1/2}$ by (15). By [10, Lem. A.3], each frontier summand is negative with magnitude at least $\frac{1}{2}w_\gamma^\Gamma$ for all large λ . Summing over the frontier band and dividing by $\sqrt{\lambda}$ gives

$$\sum_{\ell_\gamma \in [L-1, L]} w_\gamma^\Gamma \geq (1 - e^{-1}) \frac{\lambda}{\log \lambda} \cdot \frac{\log \lambda}{\sqrt{\lambda}} \cdot \frac{1}{2} (1 + o(1)) = \frac{1}{2} (1 - e^{-1}) \sqrt{\lambda} (1 + o(1)),$$

yielding $c_\Gamma \geq (1 - e^{-1})/2$. □

Remark 4.9. The factor $(1 - e^{-1}) \approx 0.632$ is the geometric ratio of frontier geodesic density (exponential regime) to frontier prime density (polynomial regime): Selberg has $(1 - e^{-1})\lambda/\log \lambda$ frontier geodesics where Riemann has $\lambda/\log \lambda$ frontier primes. This accounts for the only structural difference between the two frontier estimates.

The complete analytical and asymptotic correspondence between the arithmetic Riemann regime and the hyperbolic geometric Selberg regime is summarized in Table 1.

4.4 Conditional Reduction via v2.0

Theorem 4.10 (v2.0 Selberg-RH, conditional). *Let $M = \Gamma \backslash \mathbb{H}$ be a compact hyperbolic surface satisfying $\lambda_1(\Delta_\Gamma) \geq 1/4$. Assume the following three C2 conditions for QW_λ^Γ :*

- (C2a) QW_λ^Γ has a lower-bounded self-adjoint realization on the stated interval model;
- (C2b) for all $\lambda \geq \lambda_0(\Gamma)$ its lowest eigenvalue is simple and attained by an even minimizer $\eta_{\lambda, \Gamma}$;
- (C2c) the Fourier/Selberg transform of $\eta_{\lambda, \Gamma}$ is identified, in the relevant exact model or limit, with the completed Selberg spectral factor whose zeros encode Z_Γ .

Then all non-trivial spectral zeros of $Z_\Gamma(s)$ in the critical strip lie on $\text{Re}(s) = 1/2$. Without (C2c), Connes' mechanism only gives a statement about the zeros of the minimizer transform, not about the spectral zeros of Z_Γ .

Analytical Parameter / Quantity	Riemann Arithmetic Regime (ζ)	Selberg Geometric Regime (Z_Γ)
Fundamental counting objects	Primes $p \in \mathbb{P}$ (discrete arithmetic prime numbers)	Primitive closed geodesics $\gamma \in \mathcal{P}_\Gamma$ (length spectrum ℓ_γ)
Asymptotic density / Counting law	Prime Number Theorem: $\pi(x) \sim x / \log x$	Prime Geodesic Theorem: $\pi_\Gamma(L) \sim e^L / L$
Frontier interval definition	$p \in [\lambda/e, \lambda]$ on multiplicative scale λ	$\ell_\gamma \in [L-1, L]$ on additive scale $L = \log \lambda$
Frontier object count $\Delta\pi$	$\pi(\lambda) - \pi(\lambda/e) \sim \frac{\lambda}{\log \lambda}(1 - e^{-1})$	$\pi_\Gamma(L) - \pi_\Gamma(L-1) \sim \frac{\lambda}{\log \lambda}(1 - e^{-1})$ (asymptotically identical)
Single-element weight	$w_p = \frac{\log p}{\sqrt{p}} \sim \log \lambda \cdot \lambda^{-1/2}$	$w_\gamma^\Gamma = \frac{\ell_\gamma}{2 \sinh(\ell_\gamma/2)} \sim L e^{-L/2} =$ $\log \lambda \cdot \lambda^{-1/2}$
Frontier sum normalization	$\lambda^{-1/2} \sum_{p \in \text{frontier}} w_p \rightarrow 1 - e^{-1} \approx 0.632$	$\lambda^{-1/2} \sum_{\gamma \in \text{frontier}} w_\gamma^\Gamma \rightarrow 1 - e^{-1} \approx 0.632$
Frontier dominance constant c	$c_{\text{Riem}} \geq \frac{1}{2}$ (arithmetic parity advantage)	$c_\Gamma \geq \frac{1}{2}(1 - e^{-1}) \approx 0.316 > 0$ (Lemma 4.7)
Remainder / Volume scale	Archimedean contribution: $O(\log \lambda)$	Identity contribution: $\frac{\text{Area}(M)}{4\pi} O(L) = O(\log \lambda)$
Universal commutant (NE-B)	Obstruction holds: no universal operator commuting with prime shifts	Obstruction fails: Casimir $[\Delta_\Gamma, R_k] = 0$ on spherical convolution algebra
Operator realization	Absent / unconstructed on $L^2([-L, L])$	Laplace–Beltrami operator Δ_Γ with spectrum $\lambda_n = s_n(1 - s_n)$

Table 1: Asymptotic regime and density parameter ledger: comparison between the arithmetic Riemann framework and the hyperbolic geometric Selberg framework.

For readability, the C2 gate is recorded separately from the theorem statement. Table 2 distinguishes what is already used inside this paper from the operational ledger that remains a companion-level audit rather than a theorem-level closure. The no-exceptional-eigenvalue assumption is shown as a separate SG/NoExc gate; it is not part of C2b.

Proof sketch. The structure of the argument is identical to the v2.0 proof for Riemann.

Step 1 (pointwise even preference, including iterates). By Proposition 4.1, for every $\gamma \in \mathcal{P}_\Gamma$ and every $m \geq 1$ with $m\ell_\gamma \leq 2L$, the difference matrix $D_N^\Gamma(m\ell_\gamma/L)$ has $\lambda_{\min} < 0$. Here $m > 1$ corresponds to the m -fold iterate of γ ; the SPL applies to iterates on exactly the same footing as to primitive geodesics, since the shift length $m\ell_\gamma$ enters the SPL as a scalar parameter $r = m\ell_\gamma/L$. By Remark 4.3, iterates with $m\ell_\gamma > 2L$ contribute exactly zero and need not be handled separately. Hence each individual geodesic summand (primitive or iterated), weighted by the positive coefficient $\ell_\gamma/(2 \sinh(m\ell_\gamma/2))$, contributes negatively to the even-minus-odd difference.

Step 2 (frontier dominance). By (14) and (16), the aggregate contribution of frontier geodesics to $\Delta^\Gamma(\lambda)$ is negative of order $-c_\Gamma \sqrt{\lambda}$, while the combined contribution of non-frontier geodesics plus the archimedean term is controlled by the Selberg analogue of the Part-II remainder estimates. This is exactly the conditional estimate recorded in Lemma 4.7.

Step 3 (C2b parity/minimizer gate). Simplicity and evenness of the minimum are not proved here from generic simplicity or from a direct identification with $\lambda_1(\Delta_\Gamma)$. They are precisely condition (C2b). The Selberg/Casimir channel is strong evidence for the

Gate	Object recorded	Current use in this paper	Failure or revision signal
SPH: spherical channel	Selberg transform and radial K -bi-invariant convolution acting through $\widehat{k}(r_j)$, with Casimir/Laplace commutation.	Positive-control context for NE-B failure; it is not itself a C2 gate and does not identify individual geodesic-flow pullbacks with interval shifts.	A non-radial or full SM -shift model is used as if it were the truncated $L^2([-L, L])$ model.
C2a: interval realization	Lower-bounded self-adjoint interval model for QW_λ^Γ .	Explicit v2.0 hypothesis for applying the Connes mechanism.	The interval model is only asserted by analogy or by the spherical channel, without a stated self-adjoint realization.
C2b: parity/minimizer gate	Simple lowest eigenvalue and even minimizer for the interval realization.	Explicit v2.0 hypothesis; supported by the Selberg/Casimir channel but not derived here from the spectral gap.	The minimum is degenerate, odd, or selected only after observing the target Selberg factor.
SG/NoExc: spectral-line input	$\lambda_1(\Delta_\Gamma) \geq 1/4$ and absence of exceptional eigenvalues.	Explicit hypothesis for the strict critical-line form; the spectral zero description itself remains classical.	An exceptional eigenvalue $0 < \lambda_j < 1/4$ produces real zeros $1/2 \pm \delta_j$.
C2c: minimizer/window identification	Target window I , target source, center ν , outer gap, residual, off-window mass, and negative control.	Operational ledger now calibrated on real spectral data (Section 4.5); the Connes-minimizer identification itself remains open, so C2c stays a companion/audit gate.	The candidate has the right Rayleigh center but wrong tail or off-window mass (<code>same_rayleigh_wrong_tail</code>).
Decision layer	pass , ambiguous , or fail for each surface/window candidate.	Positive-control audit, not a proof upgrade; it localizes what remains to be checked for JST or companion work.	Missing table, same-run target choice, or absent negative controls.

Table 2: C2/minimizer-identification ledger for the Selberg positive-control reading.

right parity structure, but the present paper treats this as a stated gate. C2b is not the separate SG/NoExc spectral-gap gate.

Step 4 (Connes’ reduction). Theorem 2.1 is the *algebraic mechanism*: given ingredients (i) a lower-bounded self-adjoint operator on a symmetric interval and (ii) a simple minimum with even eigenfunction, it forces all zeros of the minimum eigenvector’s Fourier transform onto the real line. Verifying ingredients (i) and (ii) is the $C2^\Gamma$ substep, while identifying the resulting minimizer transform with the completed Selberg factor is condition (C2c). (In the three-component decomposition $RH_X = GBC_X + C2_X + \text{Hurwitz}_X$ developed in the companion work [11], in preparation, these correspond to the Lie-origin branch plus the explicit minimizer-identification gate.) For Riemann, ingredient (ii) is non-trivial; it is the content of v2.1 Proposition A6 [10]. Applying Connes reduction with QW_λ^Γ in place of QW_λ , the zeros of $\hat{\eta}_{\lambda,\Gamma}$ are real. Condition (C2c) is the additional gate that transfers this real-zero statement to the completed Selberg factor. Together with $\lambda_1(\Delta_\Gamma) \geq 1/4$, this gives the strict statement for the non-trivial Selberg spectral zeros in the critical strip under the theorem’s explicit hypotheses. $\square$

Remark 4.11 (Conditionality and consistency). Theorem 4.10 is a conditional reduction: *given* the C2 interval/minimizer gates and the separate no-exceptional-eigenvalue condition, the strict spectral-zero statement follows by the v2.0 argument. Since the classical Selberg theory already gives the spectral zero description and ties the strict line statement to the gap $\lambda_1(\Delta_\Gamma) \geq 1/4$, it verifies the framework’s conclusion independently at the correct hypothesis level. The value of the conditional reduction lies in *stress-testing the reach of the v2.0 method*: it applies in a case where a Hilbert–Pólya operator already exists, and the two routes reach the same conclusion by independent means. This agreement is evidence of consistency, not a proof upgrade.

Remark 4.12 (Even dominance as structural expectation). Condition (C2b) in Theorem 4.10 is a *structural expectation*: it is suggested by the positivity of the hyperbolic geodesic weights and by the Casimir/spherical trace-formula channel, but it is not proved here. A fully rigorous proof would require the same frontier-dominance estimates as in the Riemann Part II [10], adapted to the exponential geodesic density, and the minimizer-identification gate (C2c). Lemma 4.7 provides only the conditional frontier input. For Selberg, the structural expectation is additionally supported by the independent operator-theoretic proof (Theorem 3.2), which confirms the conclusion without using even dominance at all.

4.5 C2c Window Ledger with Real Spectral Data

The C2c gate of Table 2 asks for an operational window certificate. For a candidate vector ψ , a spectral window I with center ν , and the relevant self-adjoint operator D (here $D = \Delta_\Gamma$, resp. its cuspidal restriction), the spectral theorem gives

$$\|(\text{Id} - P_I)\psi\| \leq \frac{\|(D - \nu)\psi\|}{\text{dist}(\nu, \sigma(D) \setminus I)}, \quad P_I = \mathbf{1}_I(D). \quad (18)$$

This subsection reports a calibration of this certificate on *real, independently computed* spectra, replacing the internal toy spectrum of the earlier project audit. Two positive-control surfaces are used:

- the *Bolza surface* (compact, genus 2): the first ten positive Laplace eigenvalues with multiplicities from the certified computation of Strohmaier–Uski [24]. Compactness

makes this the lemma-conformal main case; the method of [24] finds *all* eigenvalues below the cutoff, so the finite list is complete below $\lambda_{\max} \approx 30.83$;

- the *modular surface* $\mathrm{PSL}(2, \mathbb{Z}) \backslash \mathbb{H}$ (level 1): the first eight cuspidal spectral parameters r ($\lambda = 1/4 + r^2$) from the rigorous Maass-form computations of Seymour–Howell [23] and the rigorous database described in [17]. This surface is *not* compact: all corresponding rows refer exclusively to the self-adjoint restriction Δ_{cusp} on L^2_{cusp} , and the continuous Eisenstein spectrum $[1/4, \infty)$ of the full Laplacian lies outside the ledger scope.

The certificate (18) is evaluated as a *finite-spectrum proxy with an explicit tail budget*. For multiplier candidates $h(D)\psi_0$, the spectral mass above the data cutoff Λ is bounded by numerically integrated tail budgets $T_0 \geq \sum_{\lambda > \Lambda} \text{mult} \cdot |h(\lambda)|^2$ and $T_2 \geq \sum_{\lambda > \Lambda} \text{mult} \cdot |h(\lambda)|^2 (\lambda - \nu)^2$, using a documented counting-density assumption above the cutoff (twice the Weyl main term $\text{Area}/4\pi$: $\rho = 2$ for Bolza, $\rho = 1/6$ for the cuspidal modular case). The reported residual/gap and off-window columns (both norm quantities, matching the norm form of the certificate) are worst-case adjusted by these budgets, and a bracketing check verifies $\Lambda - \nu > \text{dist}(\nu, \sigma(D) \setminus I)$ for every window, so eigenvalues above the cutoff cannot shrink the reported gap. The audit thresholds (**pass** ≤ 0.25 , **ambiguous** ≤ 0.5) are a fixed convention of the audit, not part of the spectral-theorem bound.

Candidates are grouped into four control classes: *positive* controls (window eigenvectors and tuned window multipliers $h(D)$), *radial miscandidates* (genuine functions of D — hence exactly commuting — but miscentred or badly tuned), *mathematical negatives* (adversarial spectral distributions, including the mandatory **same_rayleigh_wrong_tail** class), and *bookkeeping ablations* (structure breakers such as permuted spectral bookkeeping, which test the implementation rather than the mathematical claim). For candidates of the form $h(D)$ the commutator leakage term vanishes identically by functional calculus; this is a *class* statement, not a verified property of any external minimizer candidate.

The outcome, recorded in Table 3, is exactly the calibration pattern required by the C2c row of Table 2: all positive controls pass also after tail adjustment (the tuned Gaussian budgets are numerically negligible; the heat kernel restricted to L^2_0 — admissible since L^2_0 is Δ_Γ -invariant — remains a borderline pass at ratio 0.248); the *untuned* heat kernel $e^{-\Delta_\Gamma}$ fails because it concentrates on the constant ground state $\lambda_0 = 0$ rather than on the target window (a correct diagnosis, not a false alarm); the two genuine-but-mistuned $h(D)$ candidates fail *inside* the commuting class, so commutation alone does not rescue a badly built window; and the mandatory negative class **same_rayleigh_wrong_tail** fails on real data despite matching the Rayleigh center exactly. The ledger script and the full result files (including the T_0/T_2 tail columns and the bracketing check) are available in the project repository.¹

Remark 4.13 (No claim upgrade). This ledger operationalizes and calibrates the C2c *certificate* on real spectra; it does not close the C2c *gate*. What remains open is the core identification: constructing the Connes minimizer $\eta_{\lambda, \Gamma}$ itself as a vector, verifying that it lies in the $h(\Delta_\Gamma)$ class (so that the class statement on commutator leakage applies to it), and passing it through this ledger. Until then C2c retains companion/audit status, the counting-density bound above the cutoff remains a documented assumption (twice the Weyl main term) rather than a cited remainder bound, and no conclusion of this paper is upgraded.

¹<https://github.com/research-line/functional-stability-theory/tree/main/masters/seiberg>

Window	Candidate	Class	Res./gap	Off-win.	Decision
Bolza λ_1 (mult. 3)	window eigenspace	positive	0.000	0.000	pass
	tuned Gaussian multiplier	positive	0.002	0.002	pass
	untuned heat kernel $e^{-\Delta_\Gamma}$	miscand.	2.533	0.999	fail
	heat kernel on L_0^2	positive	0.248	0.246	pass
	wide Gaussian ($\sigma=5$)	miscand.	1.123	0.758	fail
	miscentred bandpass (at 10)	miscand.	2.912	1.000	fail
	same-Rayleigh wrong-tail	negative	1.592	1.000	fail
	false center	negative	1.000	1.000	fail
	broken radial commutation	ablation	7.188	1.000	fail
	permuted spectrum	ablation	16.003	1.000	fail
Bolza λ_8 (isolated)	window eigenspace	positive	0.000	0.000	pass
	tuned Gaussian multiplier	positive	0.003	0.003	pass
	same-Rayleigh wrong-tail	negative	1.398	1.000	fail
Modular λ_2 (cuspidal)	window eigenvector	positive	0.000	0.000	pass
	tuned Gaussian multiplier	positive	0.000	0.000	pass
	same-Rayleigh wrong-tail	negative	1.172	1.000	fail

Table 3: C2c window ledger on real spectra (Rev. 2; residual/gap and off-window norm values are tail-adjusted). Windows: Bolza λ_1 (multiplicity 3), Bolza λ_8 (isolated), modular surface λ_2 (cuspidal; Δ_{cusp} scope).

4.6 NE-A for the Selberg Setting

Define the Selberg prime shift multiplier

$$M^\Gamma(\xi) = -2 \operatorname{Re} \left[\frac{Z'_\Gamma}{Z_\Gamma} \left(\frac{1}{2} + i\xi \right) \right]. \quad (19)$$

Proposition 4.14 (NE-A for Selberg). *The set $\{\xi \in \mathbb{R} : M^\Gamma(\xi) < 0\}$ has positive Lebesgue measure.*

Proof sketch. By the Hadamard product representation of the Selberg zeta function on a compact hyperbolic surface ([16, Ch. 10]), one has

$$\begin{aligned} \frac{Z'_\Gamma}{Z_\Gamma}(s) &= b_\Gamma + \sum_{\rho} \left[\frac{1}{s - \rho} + \frac{1}{\rho} \right] \\ &\quad + (\text{contributions from trivial zeros and the topological factor}), \end{aligned} \quad (20)$$

where ρ runs over the non-trivial zeros (on $\operatorname{Re}(s) = 1/2$ under Theorem 3.2) and $b_\Gamma \in \mathbb{R}$ is a constant depending on Γ .

Setting $s = 1/2 + i\xi$ and taking real parts, each non-trivial zero $\rho_j = 1/2 + ir_j$ contributes

$$\operatorname{Re} \left[\frac{1}{i(\xi - r_j)} + \frac{1}{1/2 + ir_j} \right] = \frac{1/2}{1/4 + r_j^2} > 0. \quad (21)$$

Thus the non-trivial zeros produce a *positive constant* contribution to $\operatorname{Re}[Z'_\Gamma/Z_\Gamma(1/2 + i\xi)]$; they do not create sign oscillations in $M^\Gamma(\xi)$.

The sign oscillations arise from the trivial zeros (at $s = -k$, $k \geq 0$) and the topological factor, whose contributions to $\text{Re}[Z'_\Gamma/Z_\Gamma]$ are of the form $\sum_{k \geq 0} c_k / [(\xi^2 + (k + 1/2)^2)]$ with $c_k \in \mathbb{R}$ of alternating sign. Combining with the explicit-formula identity term of the Selberg trace formula (7), one finds by a direct application with a suitable test function (localizing in ξ) that $M^\Gamma(\xi)$ takes both strictly positive and strictly negative values on sets of positive Lebesgue measure. The argument is the direct analogue of [10, Lem. A.3]; the only substitution is $\tanh(\pi r)$ in place of $\Gamma'/\Gamma(1/4 + ir/2)$ in the archimedean kernel. $\square$

NE-A therefore transfers. The prime shift multiplier is oscillatory in both settings. In the Selberg case the sign changes come from the trivial/topological/identity terms in the explicit formula, not from the critical-line zeros themselves.

5 NE-B Fails for Selberg: The Spherical Casimir Channel

Here we come to the substantive difference between Selberg and Riemann.

5.1 Statement of the Commutation Proposition

Proposition 5.1 (Casimir commutation in the spherical Selberg channel). *Let $G = \text{PSL}(2, \mathbb{R})$ and $K = \text{SO}(2)$, so $M = \Gamma \backslash G/K$. Let R_k be the radial convolution operator on $L^2(\Gamma \backslash G/K)$ associated with a K -bi-invariant test kernel k . Then, on the common invariant domain,*

$$[\Delta_\Gamma, R_k] = 0. \quad (22)$$

Consequently the Selberg trace-formula channel has a non-trivial Casimir commutant. In this precise spherical sense, the Selberg analogue of the NE-B obstruction is absent.

5.2 Spherical Radial Convolution Operators

The objects that v2.0 manipulates directly are shifts on the *logarithmic interval* $[-L, L]$. In the Riemann setting, the variable t corresponds to $\log x$ in the idele class space (in Connes' adèle framework). In the Selberg setting, the analogous length variable in the spherical trace formula channel is implemented via radial convolution kernels.

For a compactly supported K -bi-invariant kernel $k \in C_c^\infty(K \backslash G/K)$, set

$$(R_k f)(x) = \int_G k(g) f(xg) \, dg, \\ f \in L^2(\Gamma \backslash G/K),$$

on the automorphic quotient whenever the integral is defined. These radial operators represent the spectral side of the Selberg trace formula: under the Selberg/Harish–Chandra transform, R_k is diagonalized on Laplace eigenfunctions:

$$\Delta_\Gamma \psi_j = (1/4 + r_j^2) \psi_j, \quad R_k \psi_j = \hat{k}(r_j) \psi_j.$$

The trace formula writes the geodesic length sum in precisely this spherical test-function channel. This is the true Selberg analogue of the commutator test: not individual pullbacks of the geodesic flow on SM , but the radial convolution operators whose Selberg transforms see the spectral parameters r_j .

5.3 Proof of Proposition 5.1

Proof. We prove (22) by identifying both Δ_Γ and R_k within the spherical convolution algebra on $\Gamma \backslash G/K$.

Step 1 (Casimir centrality). The Laplace–Beltrami operator Δ_Γ is the image of the Casimir element of the universal enveloping algebra $\mathcal{U}(\mathfrak{sl}_2)$. This element belongs to the center and therefore commutes with the regular G -action.

Step 2 (Radial convolution). The operator R_k is an average of right translations with bi- K -invariant kernel k . Since Δ_Γ commutes with every such translation, Δ_Γ also commutes with their integral R_k .

Step 3 (Joint diagonalization). Equivalently, the Laplace eigenbasis diagonalizes both operators:

$$\Delta_\Gamma \psi_j = (1/4 + r_j^2) \psi_j, \quad R_k \psi_j = \hat{k}(r_j) \psi_j.$$

Hence $[\Delta_\Gamma, R_k] = 0$ on the spherical spectral channel. Consequently, Δ_Γ serves as a non-trivial universal commutant for the relevant Selberg test-function algebra. $\square$

Remark 5.2 (No individual phase-eigenvalue claim). The paper does not use the stronger and generally unsafe assertion that an individual geodesic-flow element satisfies $T_t^\Gamma \psi_j = e^{itr_j} \psi_j$ for arbitrary Laplace eigenfunctions. The safe statement is the spherical one: radial convolution operators are diagonalized by the Selberg/Harish–Chandra transform with scalar multiplier $\hat{k}(r_j)$.

Remark 5.3 (Why this is possible for Selberg but not for Riemann). The argument depends on the existence of a spherical homogeneous channel $\Gamma \backslash G/K$ and its central Casimir. Primitive geodesics enter the trace formula through conjugacy classes and radial kernels, while the Casimir diagonalizes the same spectral side. Primes, by contrast, do not come with a known comparable spherical convolution algebra and central differential operator. This is what NE-B captures at the level used here: the tested Riemann shift model has no corresponding universal commutant.

5.4 Consequence for the Numerical Commutator Test

In the spherical Selberg model, repeating the numerical commutator test of [10, Section 5.3] should yield a *large* kernel for the commutator constraint matrix C_N – in fact a kernel whose dimension grows with the number of distinct Laplace eigenvalues sampled, because every polynomial in Δ_Γ is a commutant. The singular-value spectrum would show no “isolated-1-dim-null”-structure; instead many small singular values would appear, reflecting the infinite-dimensional commutant.

This is the diagnostic: *if you run the v2.0 commutator test on a genuine Hilbert–Pólya system, it detects the operator.* In the Riemann case, the tested truncation detects nothing nontrivial – and this is, by the logic of v2.0, evidence that this specific commutant route is absent.

6 Hilbert–Pólya as a Special Case of v2.0

We now articulate the meta-principle that emerges from combining the Riemann and Selberg analyses.

6.1 The Hilbert–Pólya Program Revisited

The classical Hilbert–Pólya conjecture proposes to prove RH by constructing a self-adjoint operator H on a Hilbert space whose spectrum is the set of imaginary parts $\{\gamma_k\}$ of the non-trivial Riemann zeros. Such an operator, if it exists, would give RH as an immediate corollary (self-adjointness $\Rightarrow$ real eigenvalues $\Rightarrow$ zeros on the critical line).

One hundred years of effort have produced candidate operators (Berry–Keating $H = xp$ [2], Connes’ adelic Dirac operator [5], and others) but none that provably realize the Riemann spectrum.

6.2 The v2.0 Alternative

The v2.0 program circumvents the Hilbert–Pólya requirement by establishing RH through a *quadratic-form inequality* (even dominance of QW_λ), rather than through an operator-spectral identification. The ingredients are:

- A parity decomposition of the ambient Hilbert space.
- An algebraic *pointwise* parity advantage (Shift Parity Lemma).
- An analytic *aggregation* mechanism (frontier dominance).
- An argument that no simpler route applies (NE-A, NE-B).

Crucially, *no operator is constructed*. The route is via the quadratic form directly.

6.3 Selberg as Calibration: The Spherical Operatorial Channel Exists

Theorem 4.10 shows that v2.0 conditionally reduces the strict statement for the non-trivial Selberg spectral zeros in the critical strip to the C2/minimizer gates, even dominance of QW_λ^Γ , and $\lambda_1(\Delta_\Gamma) \geq 1/4$. In this case, the spherical Casimir channel also exists (Proposition 5.1), and classical spectral theory independently confirms the conclusion. The two routes coexist:

- Selberg’s original spectral theory (unconditional spectral parametrization; strict critical line under $\lambda_1(\Delta_\Gamma) \geq 1/4$).
- The v2.0 reduction (quadratic: the C2/minimizer gates and even dominance imply the strict spectral-zero statement under the stated gap hypothesis).

Both arrive at the same conclusion by different routes once their respective hypotheses are supplied. The v2.0 method is therefore compatible with the operator-based route; it does not by itself replace the C2/minimizer identification.

6.4 Riemann as the Complementary Case

In the Riemann setting, NE-B (in the full-space conjectural form) asserts that the tested universal-commutant route is absent. If this conjecture holds, then the particular Hilbert–Pólya strategy modeled by these shifts is blocked. This is the *scientifically interesting* case: v2.0 seeks a proof route that does not start by postulating such a commutant.

6.5 The Meta-Theorem

Theorem 6.1 (v2.0 meta-principle, informal). *Let Z be a zeta-like function admitting an explicit formula of the Riemann–Weil type (with real-valued logarithmic shifts and positive weights). Then:*

- (i) *The v2.0 ingredients (Shift Parity Lemma, frontier dominance, parity decomposition of the associated Weil quadratic form) transfer to Z with modifications only in the density and weight constants.*
- (ii) *The v2.0 route to the corresponding Riemann Hypothesis is a candidate route whenever an appropriate frontier-density bound and $C2$ /minimizer gate hold.*
- (iii) *Failure of the relevant NE-B obstruction is evidence for an operatorial/spherical Hilbert–Pólya channel for Z .*
- (iv) *When such a channel exists, v2.0 is a consistency check against a classical operatorial route. When it does not, NE-B blocks this specific commutant strategy, not every imaginable Hilbert–Pólya realization.*

We formulate this as a *structural principle* rather than a formal theorem, because it depends on what one takes as the admissible class of zeta-like functions. For classical L-functions satisfying a standard functional equation and explicit formula (Selberg class, Connes’ family), these items are expected to hold only after the relevant operator class and $C2$ /minimizer-identification gates are specified.

Remark 6.2 (Refinement in the companion meta-paper). The informal meta-principle of Theorem 6.1 has been made technically precise in the companion paper [11] as the *Meta-Theorem* (Thm. thm:meta there), via the three-component decomposition

$$\text{RH}_X = \text{GBC}_X + C2_X + \text{Hurwitz}_X.$$

In that language, items (iii)–(iv) of our informal Theorem 6.1 are the statement that $C2_X$ is trivialized in the Lie-origin branch and open (equivalent to classical Hilbert–Pólya) in the arithmetic branch. The finer point noted in [11] is that $C2_X$ itself decomposes into a *parity substep* $C2_X^{\text{parity}}$ (simple ground state with even eigenvector) and an *eigenvector-realization substep* $C2_X^{\text{eigenvec}}$ (Hermite-prolate approximation convergence). In the current CORE status, the Riemann $C2/MS2$ block is closed in the CCM/Fourier branch by the Zookeeper three-lemma microcluster closure. This does not change the Selberg proof obligation: for Selberg, both $C2$ substeps are trivialized simultaneously by the Laplace–Casimir. The Selberg case is therefore a positive control for the normal form, not a dependent corollary of the Riemann closure.

6.6 Restatement: The Weil Quadratic Form Is Fundamental

The structural insight is:

The Weil quadratic form QW is the fundamental object; the Hilbert–Pólya operator is a realization that may or may not exist.

For one hundred years, the Riemann Hypothesis has been viewed through the Hilbert–Pólya lens: find the operator, prove RH. The v2.0 framework inverts this view: prove even dominance, prove RH, and investigate (as a separate question) whether a Hilbert–Pólya

operator happens to exist. In the Selberg spherical channel it does; in the tested Riemann shift model no comparable commutant is available. Both cases are captured by the same underlying framework.

7 Applications and Outlook

7.1 Generalized L -Functions

The v2.0 method applies in principle to any zeta-like function with a Riemann–Weil-type explicit formula. We expect the following:

- *Dirichlet L -functions* $L(s, \chi)$ for a Dirichlet character χ : the explicit formula is the prime summation weighted by $\chi(p)$. The Shift Parity Lemma transfers (the character introduces a phase, not a sign-change of the geometric parity structure). Frontier dominance transfers (with PNT replaced by the PNT for L -functions, Landau–Siegel bounds). NE-B: the tested commutant obstruction is expected to remain live unless the character introduces a new operatorial channel. In the current Zeta-Zoo cartography, Dirichlet is not closed merely by importing the untwisted Riemann package: it remains the Weg-D / χ -twisted CCM follow-up branch. A concrete Dirichlet write-up would make the meta-framework’s predictions operationally testable — it is the natural next step after the present Selberg consistency check.
- *Dedekind zeta functions* $\zeta_K(s)$ for a number field K : the explicit formula sums over prime ideals $\mathfrak{p}$ with weight $(\log N\mathfrak{p}) \cdot N\mathfrak{p}^{-m/2}$. The Shift Parity and frontier mechanisms transfer. NE-B: conjecturally no comparable spherical channel exists for generic fields, while special extra-symmetry cases need separate treatment.
- *Automorphic L -functions* (Hecke, $\mathrm{GL}(n)$ -cuspidal): the explicit formula transfers. A Hilbert–Pólya channel is available when the L -function has a spectral interpretation via a Laplacian or Casimir channel (which happens when the automorphic form is itself of geometric origin — e.g., Maass forms on a hyperbolic surface). Otherwise the NE-B analogue remains a separate obstruction to test.

7.2 The Ruelle Zeta Function for Axiom-A Flows

For an Axiom-A flow on a compact Riemannian manifold (a hyperbolic dynamical system), the Ruelle zeta function is defined by

$$\zeta_R(s) = \prod_{\tau} \frac{1}{1 - e^{-s\ell_{\tau}}}, \quad (23)$$

where the product is over primitive periodic orbits τ with minimal period ℓ_{τ} . Results such as [9] and subsequent Anosov-zeta work concern analytic continuation, resonances, and decay mechanisms; they should not be read here as a general critical-line theorem. The following item is therefore only an outlook problem.

Conjecture 7.1 (v2.0 Ruelle test problem). *For a hyperbolic flow with topological entropy h , it is worth testing whether a v2.0-style Weil quadratic form QW_{λ}^R admits frontier dominance and a useful $C2/\text{minimizer}$ gate. Any critical-line-type statement for ζ_R would require additional resonance-specific input and is not established by the present Selberg argument.*

The point of the conjecture is diagnostic: for generic Axiom-A flows, no Selberg-style symmetric-space Casimir is available, so the NE-B question and the C2 gate would have to be rebuilt from resonance theory rather than imported from the Selberg case.

7.3 The Ihara Zeta Function and the Discrete–Continuum Axis

The Ihara zeta function $\zeta_\Gamma(s)$ of a finite connected $(q+1)$ -regular graph Γ is defined by

$$\zeta_\Gamma(s)^{-1} = (1 - s^2)^{r-1} \det(I - sA + qs^2I), \quad (24)$$

where A is the adjacency matrix and r is the cycle rank of Γ [15, 12, 1]. The determinant formula is sometimes called the Ihara–Bass formula; the proofs by Hashimoto (1989) and Bass (1992) established it in full generality. The Ihara RH is the statement that all non-trivial poles lie on the circle $|s| = q^{-1/2}$, equivalent to Γ being a Ramanujan graph. It has been proved for explicit Ramanujan families constructed from $\mathrm{PGL}(2)$ -arithmetic [18].

In the v2.0 / Zookeeper framework, the Ihara case is *discrete–operatorial* (SGE-YES), in parallel with the *continuous–operatorial* Selberg case. The graph Laplacian $L = (q+1)I - A$ commutes with all automorphisms of Γ and hence with the edge-adjacency transfer matrix T . This makes L the discrete analogue of the hyperbolic Laplacian Δ_Γ : a universal commutant that renders NE-B vacuous for Ihara, for exactly the same structural reason as in the Selberg case.

The three-way comparison axis in the Zookeeper framework is:

Zeta function	Type	Commutant	v2.0 class
Selberg $Z_\Gamma(s)$	continuous, geometric	Δ_Γ (Laplace–Beltrami)	SGE-YES
Ihara $\zeta_\Gamma(s)$	discrete, combinatorial	$L = (q+1)I - A$ (graph Laplacian)	SGE-YES
Riemann $\zeta(s)$	arithmetic	none (NE-B holds)	SGE-NO

In both SGE-YES cases the strict RH-type statement is governed by a classical spectral bound (Selberg’s no-exceptional-eigenvalue condition $\lambda_1(\Delta_\Gamma) \geq 1/4$; Ramanujan’s $|\text{eigenvalue}| \leq 2\sqrt{q}$), and the v2.0 framework provides a *consistent conditional re-derivation* via even dominance of the corresponding Weil quadratic form. For the Riemann zeta, no such commutant exists; the v2.0 method is the only available structural route to RH.

7.4 Physical Interpretation

The Selberg setting corresponds physically to a quantum particle on a hyperbolic surface (“quantum chaos”); the Laplace eigenvalues are the energy levels. The v2.0 Shift Parity Lemma states that the parity of the eigenfunctions is biased toward even functions at the threshold. This is a statement about the parity-class distribution of quantum eigenstates.

For Axiom-A flows (billiards, chaotic cavities), the v2.0 framework predicts GUE-type statistics for the Ruelle zeros, as in the BGS (Bohigas–Giannoni–Schmit) conjecture [3]; see also [19] for related GUE statistics of zeros of principal L -functions. The v2.0 methodology may offer a structural explanation of this conjecture.

8 Zookeeper Consistency: Selberg as an SGE-YES Test

The Selberg case is also a useful calibration point for the post-Zookeeper version of the Zeta Zoo. In this language, Selberg is an SGE-YES system: the relevant operator exists classically, and the v2.0 transfer must therefore reproduce a known positive case rather than create a new assertion from scratch.

Table 4 records the five transfer slots and their occupants in the Selberg and Riemann settings.

Table 4: The five SGE-YES transfer slots for Selberg vs. Riemann. A tick (✓) indicates the slot is classically occupied; “v2.0” indicates it is established by the v2.0 framework; “hyp.” marks the extra spectral-gap hypothesis needed for the strict-line statement.

Slot	Selberg (SGE-YES)	Riemann
Operator / form	Δ_Γ (classical Casimir; commutes with spherical radial convolutions; interval-shift intertwiner not constructed) ✓	QW_λ (no classical HP operator; NE-B holds)
Defect	Casimir residual $S_\Gamma(L, I) = r_\Gamma/g_I$ (companion proposal)	Gap to even dominance (conjectural $\Rightarrow$ established by v2.0)
Approximation	Length cutoffs, truncated trace formula ✓	Galerkin truncation $D_N(r)$ (v2.0)
Gap	No-exceptional gap $\lambda_1(\Delta_\Gamma) \geq \frac{1}{4}$ (hyp.)	Even-dominance gap $\Delta(\lambda) > 0$ (v2.0)
Limit	Full Selberg trace formula [21] ✓	Full Weil explicit formula

The defect row uses the sharpened proof-note formulation: the Selberg defect should be measured as a residual-to-gap signal $S_\Gamma(L, I) = r_\Gamma(L, I)/g_I$, where g_I is the exterior Casimir spectral gap. This is a companion-level refinement of the positive control, not an additional unconditional Selberg-RH theorem.

Remark 8.1 (Galerkin vs. spectral truncation). In the Riemann case the approximation is a *Galerkin cutoff*: the infinite-dimensional quadratic form QW_λ is approximated by the finite matrix $D_N(r)$ via a cosine/sine basis of size N . For Selberg the approximation is instead a *spectral cutoff*: the geodesic sum is truncated at length $\ell_\gamma \leq 2L$, and Remark 4.3 shows this truncation is exact, not approximate. The Zookeeper *Approximation* slot therefore uses different but analogous technology in the two cases.

Thus Selberg plays a diagnostic role. If the Zookeeper normal form did not detect a positive gap here, the mapping would be wrong. Conversely, the fact that the Laplacian commutes with the geodesic shifts explains why the NE-B obstruction is absent in the Selberg spherical channel and highlights the contrast with the Riemann case: for Selberg the Hilbert–Pólya operator is present; for Riemann, the original Hilbert–Pólya commutant route is structurally blocked while the separate Zookeeper package closes the Riemann *C2/MS2* kernel in the CCM/Fourier branch. Selberg therefore calibrates the SGE-YES side of the boundary theorem; it is not an additional open blocker and not evidence that the Prime-Hub/OP5 graph problem has been solved.

Proposition 8.2 (Selberg–Zookeeper Consistency). *The v2.0 framework records the Selberg zeta function $Z_\Gamma(s)$ as an SGE-YES positive-control system at the stated hypothesis level. Specifically:*

- (i) *The C2/minimizer gates and even dominance of QW_λ^Γ , together with $\lambda_1(\Delta_\Gamma) \geq 1/4$, implies the strict statement for the non-trivial spectral zeros in the critical strip (Theorem 4.10).*
- (ii) *The spherical trace-formula channel has the Casimir commutant Δ_Γ , so the NE-B obstruction is absent in that precise sense (Proposition 5.1).*
- (iii) *The Zookeeper transfer slots are identified; the gap slot is conditional on the no-exceptional-eigenvalue hypothesis (Table 4).*
- (iv) *The v2.0 classification is consistent with Selberg’s classical spectral theory via Δ_Γ at the same hypothesis level.*

9 Conclusion

We have tested the v2.0 methodology on the Selberg zeta function. The Shift Parity Lemma, frontier dominance, the Weil quadratic form, and NE-A transfer to the Selberg setting only with the guardrails made explicit above: frontier dominance is conditional on the Selberg remainder bounds, and the strict v2.0 reduction for the non-trivial spectral zeros in the critical strip needs the C2/minimizer-identification gate. With these hypotheses, even dominance of the Selberg Weil quadratic form QW_λ^Γ , together with $\lambda_1(\Delta_\Gamma) \geq 1/4$, is consistent with the classical spectral description.

The decisive structural finding is that *the NE-B obstruction is absent in the Selberg spherical channel*: the Laplace–Beltrami/Casimir operator Δ_Γ commutes with radial convolution operators. This is the mark of a genuine operatorial trace-formula channel — and the v2.0 commutator test, run on such a system, should detect it.

Combining the Riemann and Selberg pictures, we formulate a meta-principle: the v2.0 framework supplies a common quadratic-form language, while the Hilbert–Pólya program is an additional operatorial realization. Selberg has such a realization in the spherical Casimir channel. In the Riemann case the tested NE-B obstruction blocks this specific commutant route; any v2.0 proof must therefore pass through the quadratic form route plus its own C2 gates.

Principle. *The Weil quadratic form is the common test object. An operatorial Hilbert–Pólya channel, when present, is extra structure.*

A century of thought has identified the operator as the heart of Riemann’s conjecture. The v2.0 framework suggests instead that the quadratic-form inequality is the heart, and the operator, where it exists, is its geometric by-product.

AI Disclosure

Statement on the Use of AI Tools

This manuscript was prepared with extensive assistance from AI-based language models (Claude, Anthropic; Gemini, Google; ChatGPT, OpenAI). AI tools were used across multiple stages of the research and writing process, including conceptual discussion, literature search, structuring, drafting support, suggestions for analysis and synthesis, critical counter-checking, language revision, translation, citation and source work, technical formatting, L^AT_EX typesetting, and compilation support.

For the computational components of this work — in particular the C2c window-ledger scripts and their evaluation runs (Section 4.5) — AI tools were also used to create, adapt, rewrite, document, and check Python scripts, and to support the planning, execution, and supervision of computational runs. This included the further development of existing scripts, the preparation of traceable outputs, the evaluation of intermediate results, reanalyses, and the documentation of tests, procedures, and results.

As part of quality assurance, several complementary procedures were used: model cross-checks, multi-stage review pipelines, explicitly falsification-oriented reviews, source and citation checks, linguistic cross-reading, reanalyses, reproducible scripts, stored result files, test runs, hash comparisons, repeated executions, and a folder-based research pipeline with project guidelines, predefined workflows, documentation rules, and documented correction steps.

Model and Workflow Assignment.

In the research work to date, simpler tasks were typically handled with Claude Opus/Sonnet 4.x, GPT-5.4, and Gemini 3.x Pro; Gemini was also used in particular for translation work. More demanding tasks were carried out with Claude Opus 4.6/4.7, Claude Fable 5, and GPT-5.5. Claude Opus was often used as a dialogic research partner for developing and advancing the research process, whereas GPT-5.5 (via the Codex toolchain) was used prominently for reviews, audits, counter-checks, and critical reanalyses — including the adversarial counter-review of the real-data window ledger reported in Section 4.5. The real-data ledger run and its integration into the present version were carried out with Claude Fable 5. Where applicable, all listed model families were also used for L^AT_EX typesetting, technical formatting, compilation, and error diagnosis. Methodologically, the work involved iterative loops, model cross-checks, inter-chat exchanges between Claude-, Gemini-, and GPT-based systems, and a multi-stage seven-step review procedure with an explicit falsification agent followed by a repair or reconstruction attempt. The human author served as the memory, connective and driving element, reflective instance, and orchestrator of the overall process; he also often contributed solutions through the attempt to understand the open problems himself.

The author determined the research question, methodological direction, central claims, evaluation of sources, disciplinary interpretation, and conclusions. Full scholarly and substantive responsibility for this article rests with the author, Lukas Geiger.

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